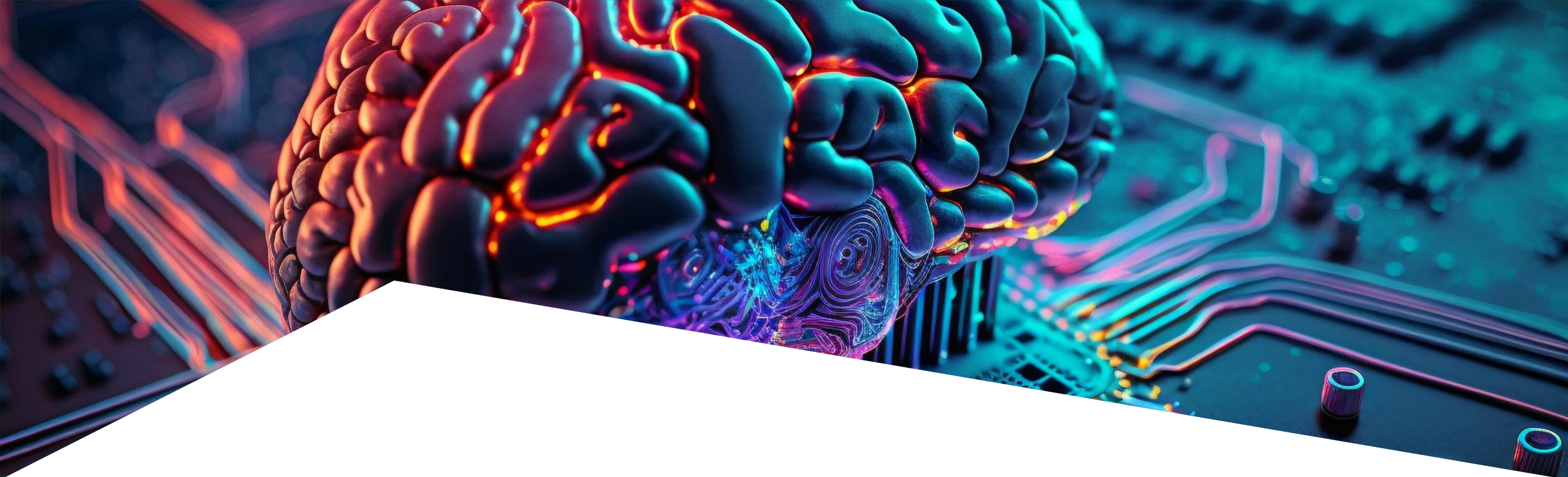




Bringing AI To the Edge

EdgeAI & DEEPCRAFT™ Studio

Gioele Mombelli – Infineon Technologies Milan (Italy)

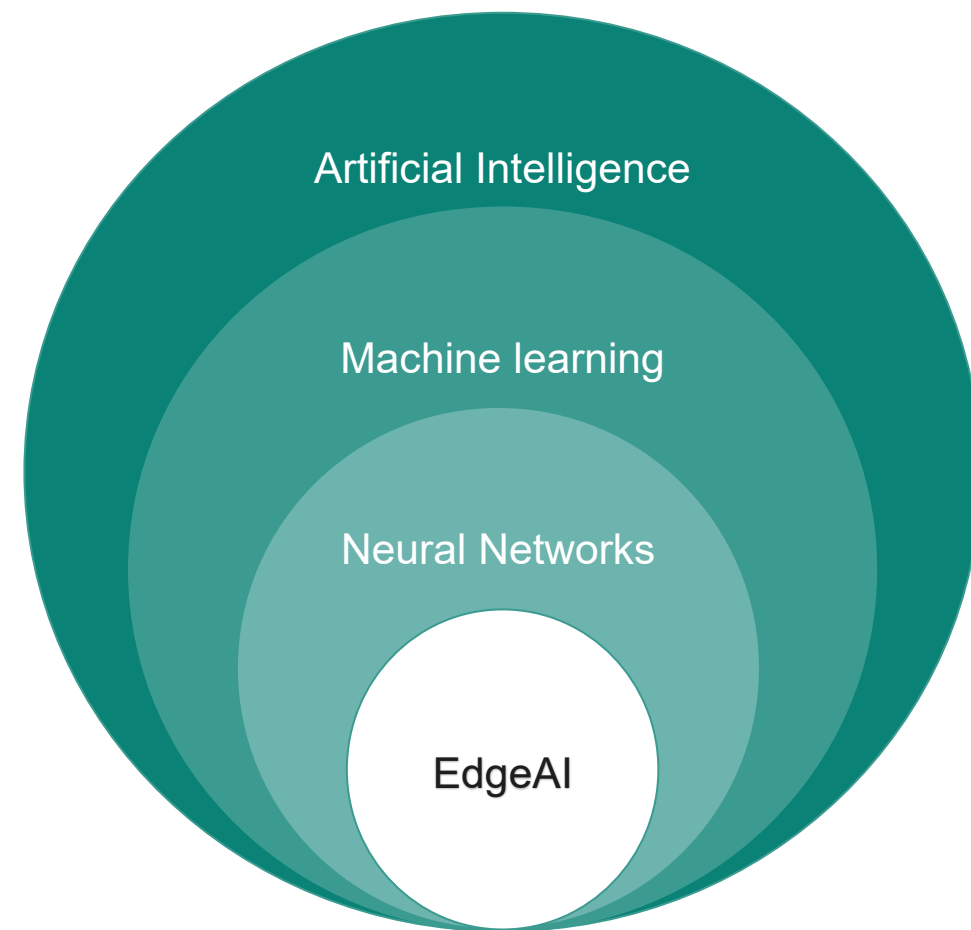


Introduction to EdgeAI and Machine Learning

Artificial Intelligence, ML and EdgeAI

Artificial Intelligence (AI) is the broad field of building machines that **simulate human intelligence**.

- **Machine Learning (ML)** is a subfield of AI that allows systems to learn and improve from data without being explicitly programmed.
- A key component of ML is **Neural Networks (NNs)**, computational models inspired by the human brain that are particularly powerful at recognizing patterns and making decisions.
- **Edge AI** takes this a step further by deploying these Neural Networks directly onto resource-constrained devices.



The concept of learning

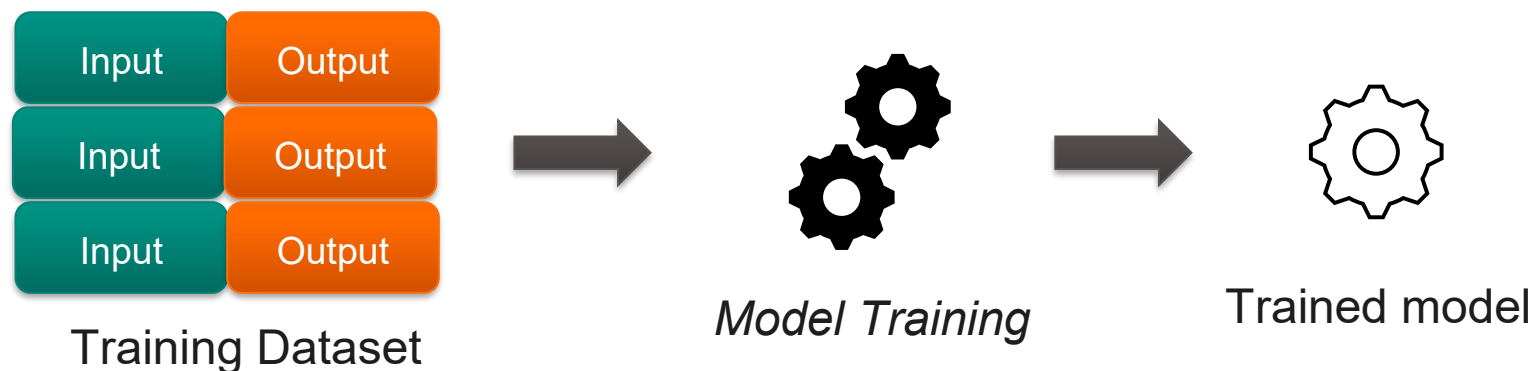
Machine Learning is based on the idea that **computers are able to *learn themselves* the algorithms to solve certain problems.**

Basic assumption of Machine Learning: it is possible to make predictions or assertions on current data based on the observation of past or similar data.



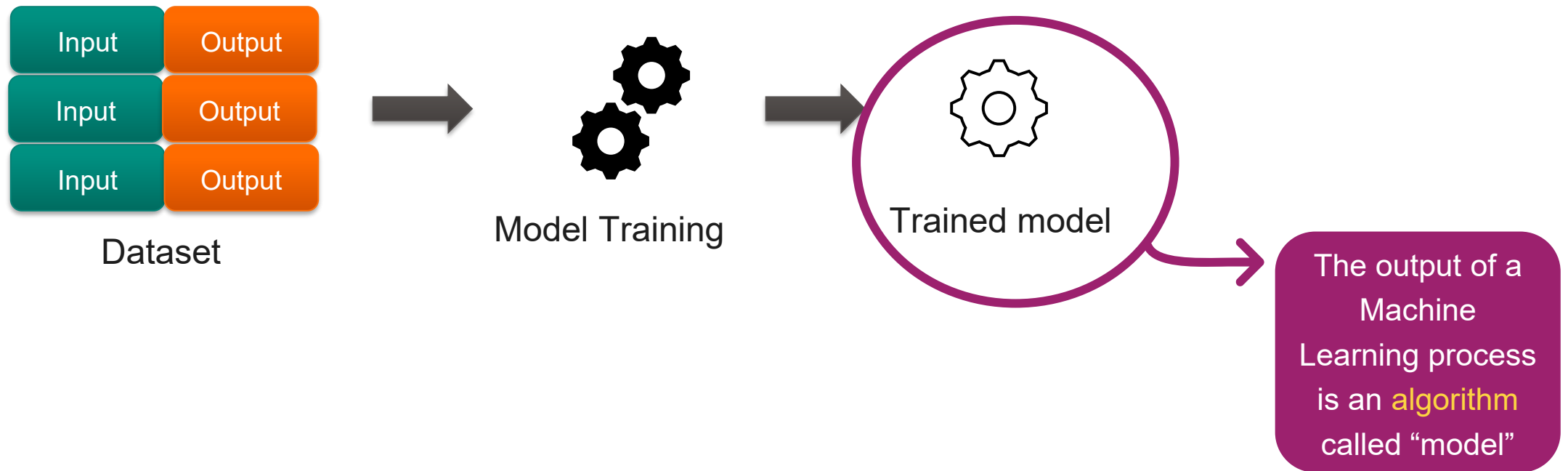
Machine Learning means learning from data

- **Machine Learning** is a subfield of AI which allows us to **build algorithms** (called «**models**») without having to **explicitly write the instructions** to solve the task.
- **How does it work:** starting from large datasets of examples composed by pairs [input; output], **machine learning** automatically **finds an algorithm** that can solve the task.
- This is achieved because models «learn» the rules to associate inputs to the correct outputs through a procedure called «**model training**».

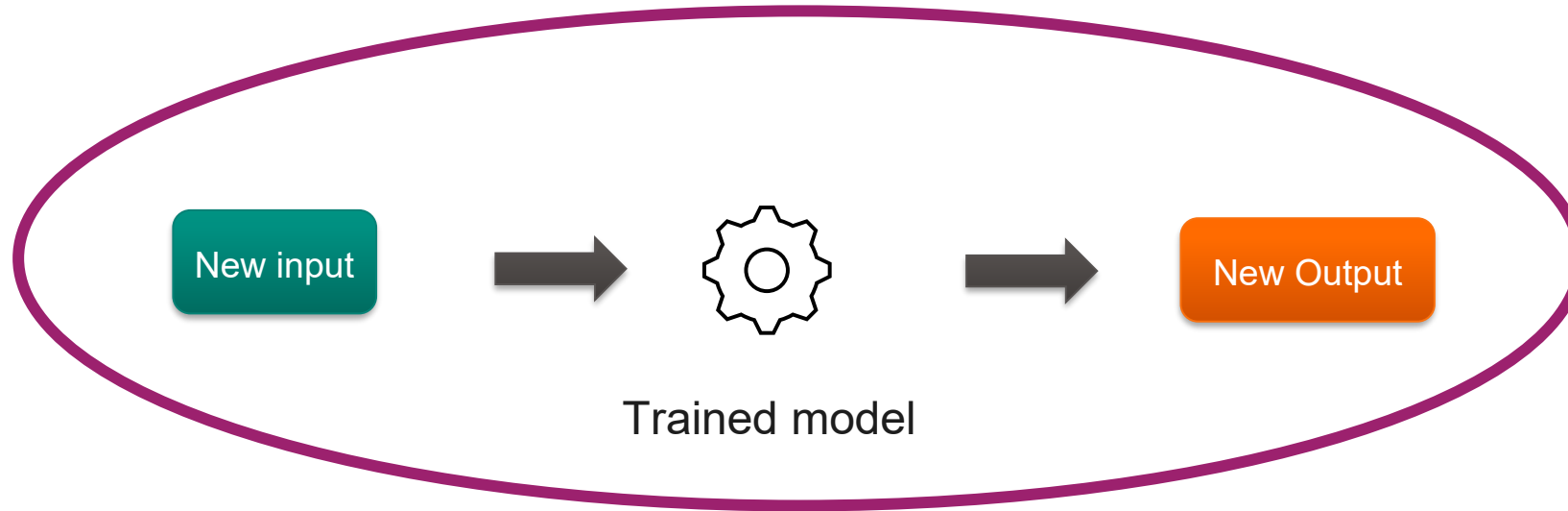


Machine Learning process

Machine Learning means learning from data



Machine Learning means learning from data



A trained model provides consistent results by associating coherent outputs to new, unseen inputs. This is called “**inference**” or “**prediction**”.

What can a Machine Learning model do?

Discriminative AI

Regression

Problem: Estimate a **precise numerical value** by processing historical data and/or real-time sensor data.

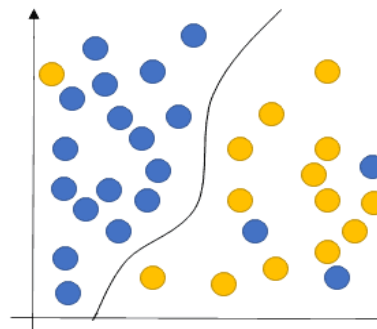


Examples:

- **Temperature Estimation** in motors and batteries using motor control/BMS data.
- **Remaining Useful Life** estimation for mechanical components.
- **Blood Pressure Estimation** from heartbeats and other biometric signals.

Classification

Problem: Assign data to a **set of discrete classes or categories**.



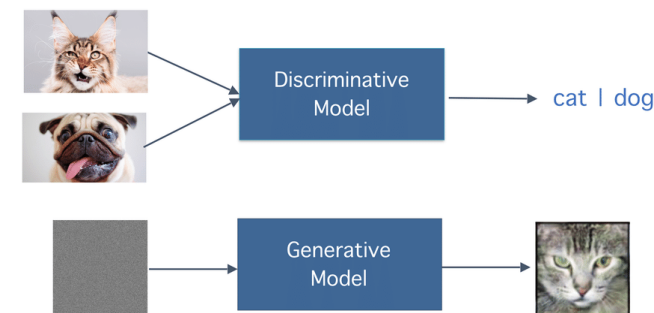
Examples:

- **Keyword Spotting** in speech and audio data.
- **Anomaly Classification** to detect if a fault is happening in industrial machineries.
- **Gesture Recognition** by processing radar or accelerometer data.

Generative AI

Generation

Problem: Generate new data samples which are coherent with the training dataset distribution.



Examples:

- **Language Models** able to generate coherent textual responses to queries;
- **Image Generation** starting from textual prompts;
- **Super-Resolution of sensor data** by performing ML-based upsampling and reconstruction.

Neural Networks are at the basis of State-Of-The-Art EdgeAI

A lot of different **Machine Learning models** have been designed in history. Starting from simple linear models all the way to complex statistical models, each one has strengths and downsides.

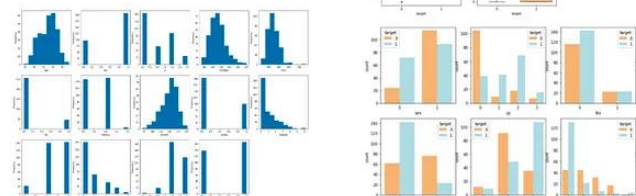
Neural Networks are a special type of ML model!

Among all the models, EdgeAI today focuses on **Artificial Neural Networks** and on how to run them on constrained devices such as MCUs.

Machine Learning Algorithms - Classification

Exploratory Data Analysis (EDA)

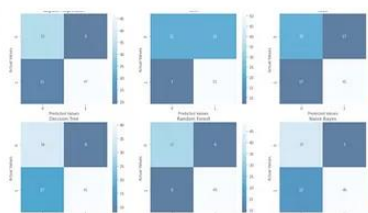
- 1) Histogram: `df.plot(kind = 'hist')`
- 2) Box Plot: `sns.boxplot()`
- 3) Grouped Bar Chart: `sns.countplot()`



Model Evaluation

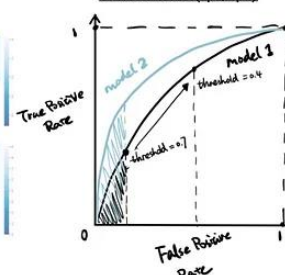
Confusion Matrix

`confusion_matrix(y_test, y_pred)`

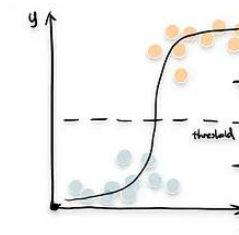


ROC & AUC

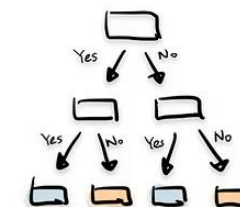
`metrics.auc(fpr, tpr)`



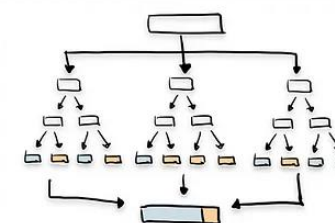
Logistic Regression



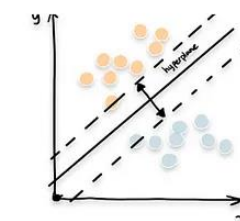
Decision Tree



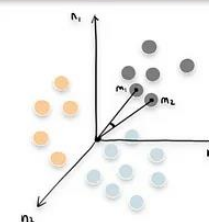
Random Forest



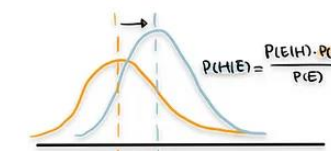
Support Vector Machine



K Nearest Neighbour



Naive Bayes



Bring neural networks to the edge

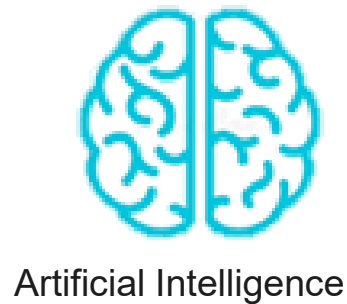
Neural Networks (NN) are incredibly powerful algorithms, but this makes them hungry for computational power and memory space. → **Microcontrollers seems not suitable for their execution!**



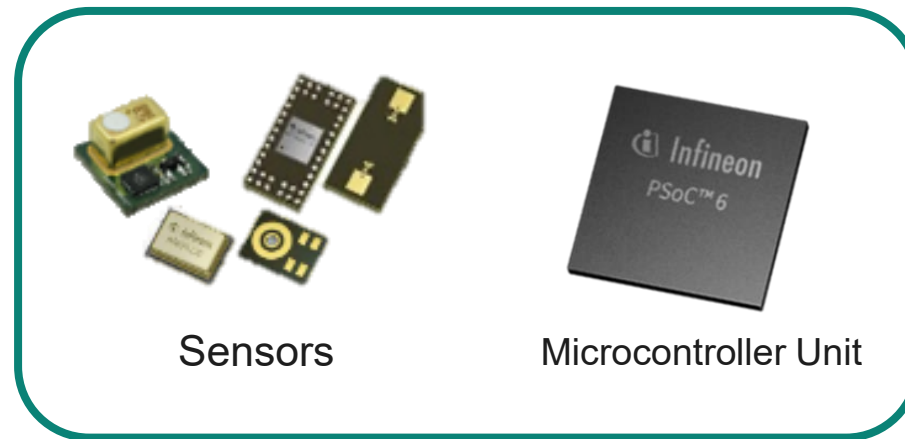
What if NN power could be condensed and compressed to run also in resource-constrained devices?

EdgeAI – Optimizing intelligent computation to the extreme

- **EdgeAI** is a young research field aimed at investigating techniques for **executing complex artificial intelligence algorithms on ultra-low power devices**.



+



- **Question:** why would we need or want to do this?

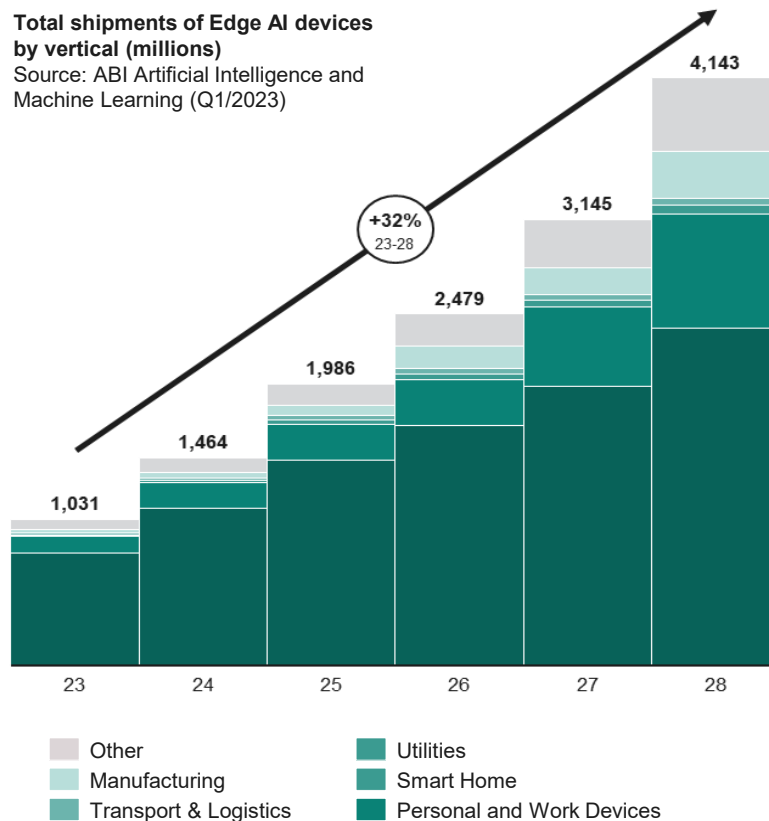
Edge AI is a key enabler of IoT and beyond, offering significant market opportunity for intelligent devices

Edge AI models run on devices at the edge of a network. Close to the source of data. Available on MCUs.

Edge AI is growing

Total shipments of Edge AI devices by vertical (millions)

Source: ABI Artificial Intelligence and Machine Learning (Q1/2023)



Benefits of Edge AI

Low latency and deterministic response: faster system response times and lower latency vs. cloud-based analytics

Improved Security & Data privacy: Data never has to leave your device, thus increasing data privacy and security

Increased system reliability: not reliant on a network that may suffer connection or bandwidth issues

Low power consumption: no streaming data leads to less power consumption

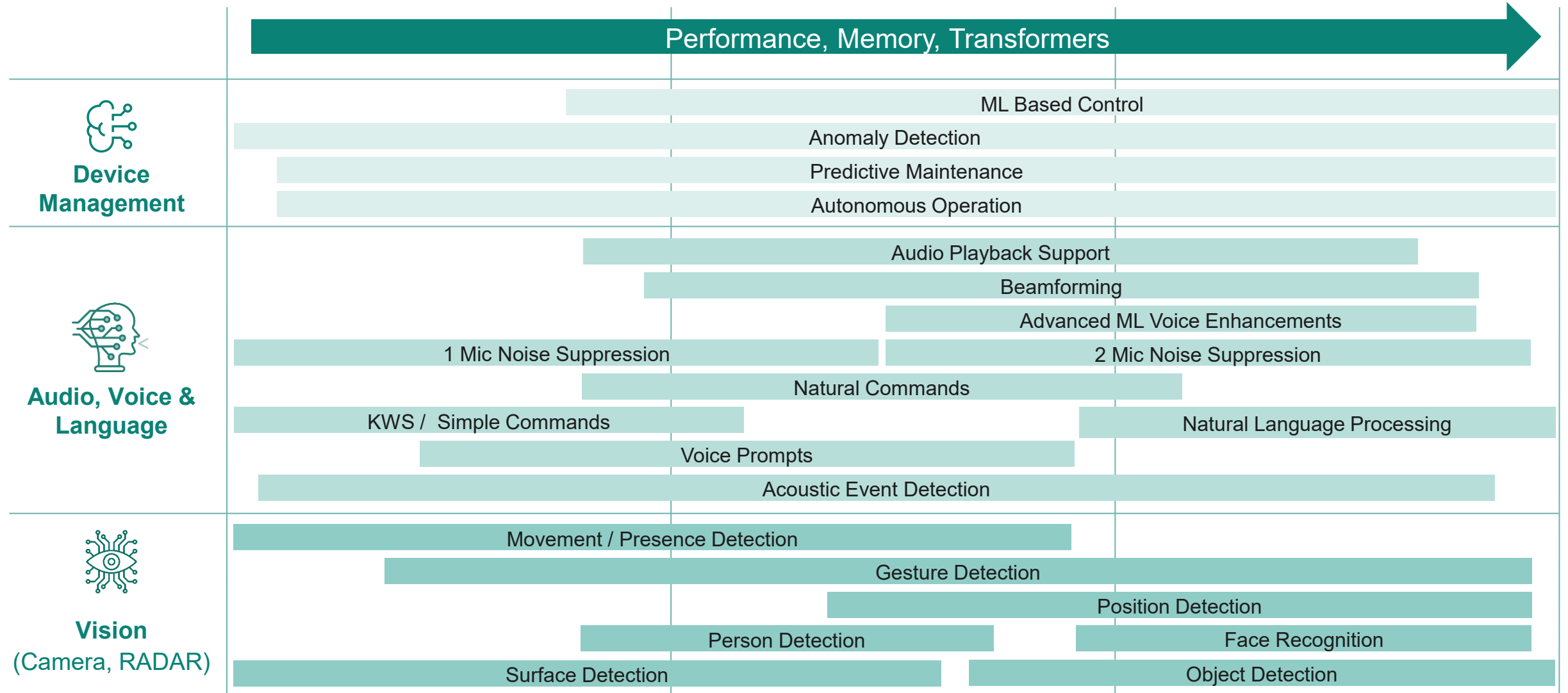
Small memory requirement: reduced memory size per processor enables smarter IoT Devices

Reduced system cost: saving cloud cost



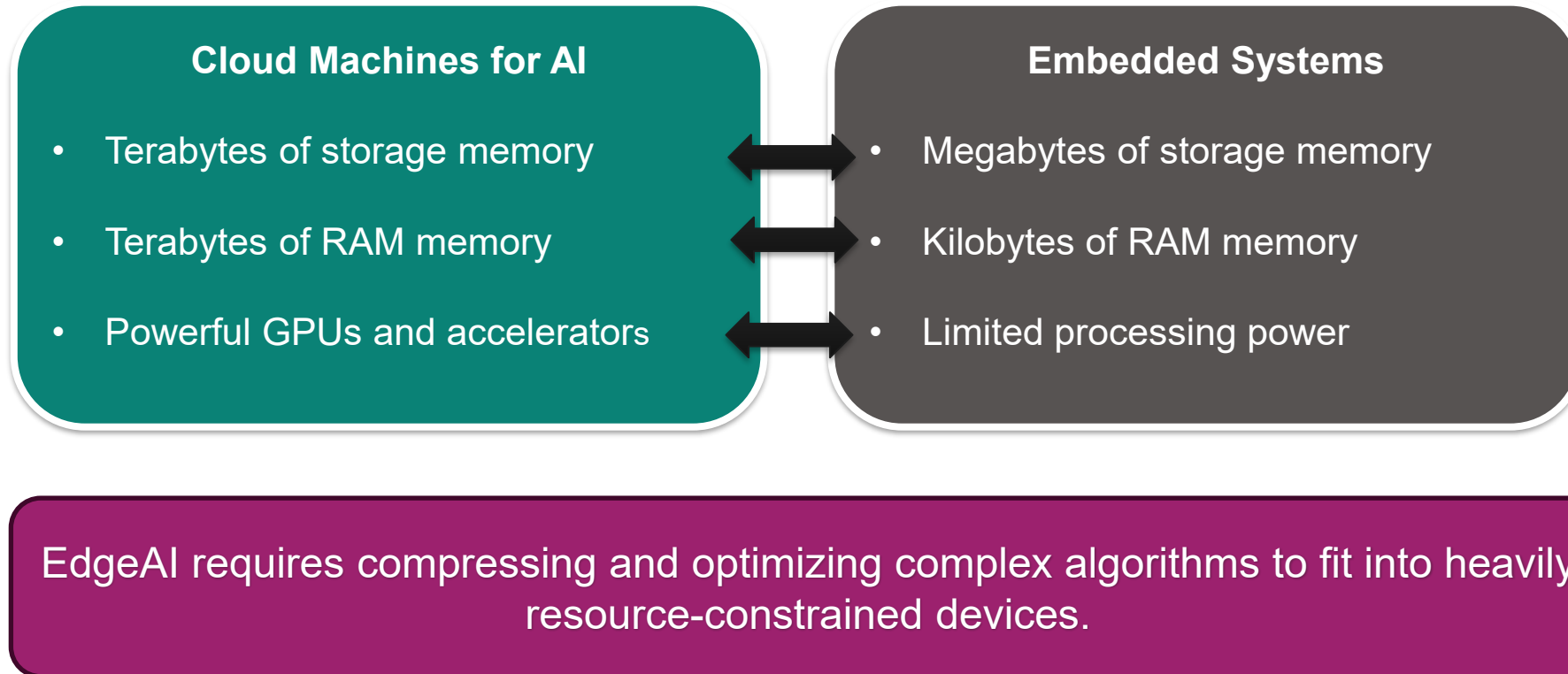
Edge AI delivers enhanced product intelligence

Enabling AI/ML use cases for Next Gen Devices



The challenge of EdgeAI

Traditional AI machines are deeply different from Embedded Systems...



We need to shrink our models as much as possible!

- Although Neural Networks are not the only ML models that can be executed to the edge, we will focus on them because of the great variety of tasks that they can solve.
- However, their complexity is a **huge obstacle** in porting them to edge devices...

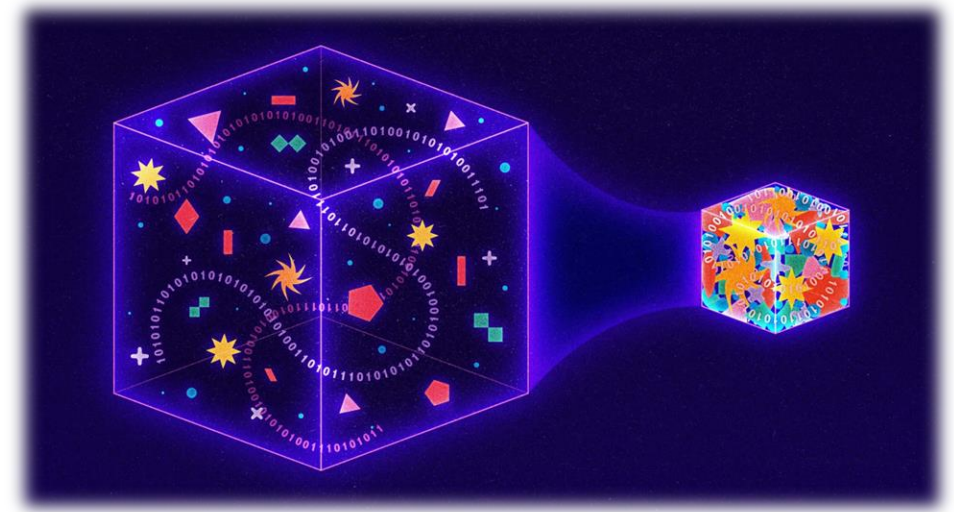
	GPT-1	GPT-2	GPT-3
Parameters	117 million	1.5 billion	175 billion
Decoded Layers	12	48	96
Hidden Layer	768	1600	12288
Context Token size	512	1024	2048

- Can you compute how much space does it take to save GPT-3? Consider that one parameter is a 32-bit value, stored in 4 bytes of memory...

Tackle the problem from two different perspectives



Study ways to build smaller and smaller effective models in advance...



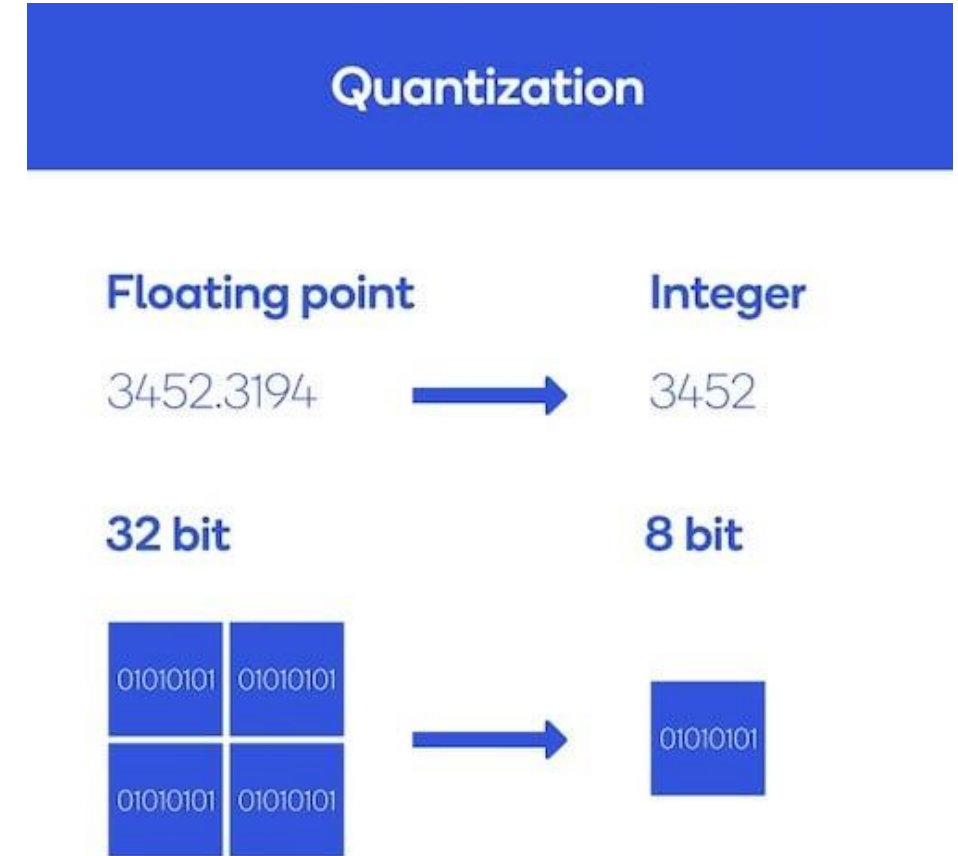
... Apply techniques for compressing and optimizing your already trained models, like model quantization.

Model quantization

- Each parameter w_n inside a neural network is usually a **floating point number** requiring **32-bit** for storage.
- What if we just **approximate floating point weights to integer values**?
- Integer values can be stored in **16 bits** or even **8 bits**, providing reduction up to 4x in model size.

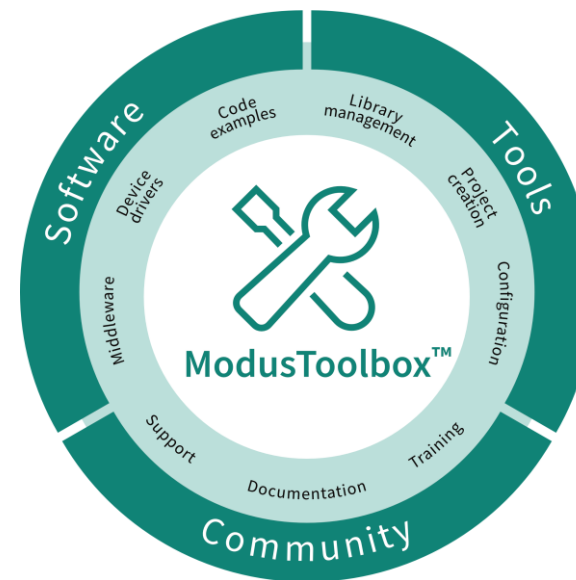
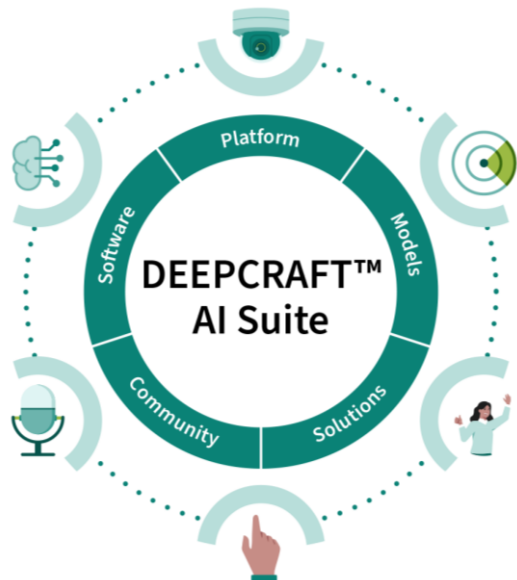
Advantages: lower storage and RAM consumption, faster execution, lower power consumption

Drawback: performance degradation; **need to quantize also input data.**



How can we practically approach EdgeAI?

- Several frameworks exist to help ML engineers approach EdgeAI.
- Infineon supports Tensorflow, Keras and Pytorch frameworks.
- Infineon also offers tools and frameworks to convert, quantize, profile and optimize models, like DEEPCRAFT™ Software.



DEEPCRAFT™ Software

Comprehensive EdgeAI Software

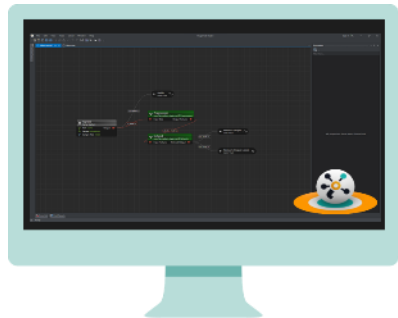
DEEPCRAFT™ AI Suite is your fast-track to intelligent, low-power Edge AI products



DEEPCRAFT™ products provide **innovative, trustworthy, convenient and green AI** software solutions for customers. Our **portfolio** offers **end-to-end Edge AI** and **best-in-class system performance in combination with Infineon Hardware**.

Platform & Tools

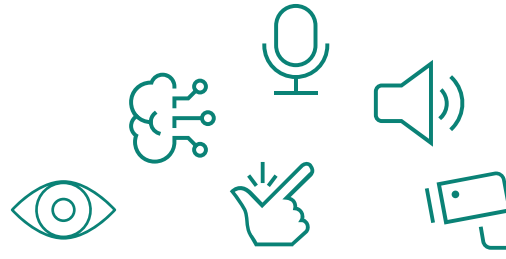
For data collection & pre-processing, model training, model conversion & deployment.



DEEPCRAFT™ Studio
20+ DEEPCRAFT™ Studio Accelerators
DEEPCRAFT™ Model Converter

Solutions

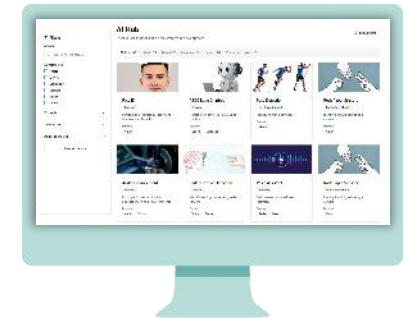
Collection of high value tools, models and configurators to add AI-based capabilities. Range across sensor and data types, from off-the-shelf to customizable solutions.



DEEPCRAFT™ Ready Models
DEEPCRAFT™ Audio Enhancement
DEEPCRAFT™ Voice Assistant

DEEPCRAFT™ AI Hub

Find the DEEPCRAFT™ Studio platform, plus the models, demos, solutions, tools and more that you need all in one centralized hub.

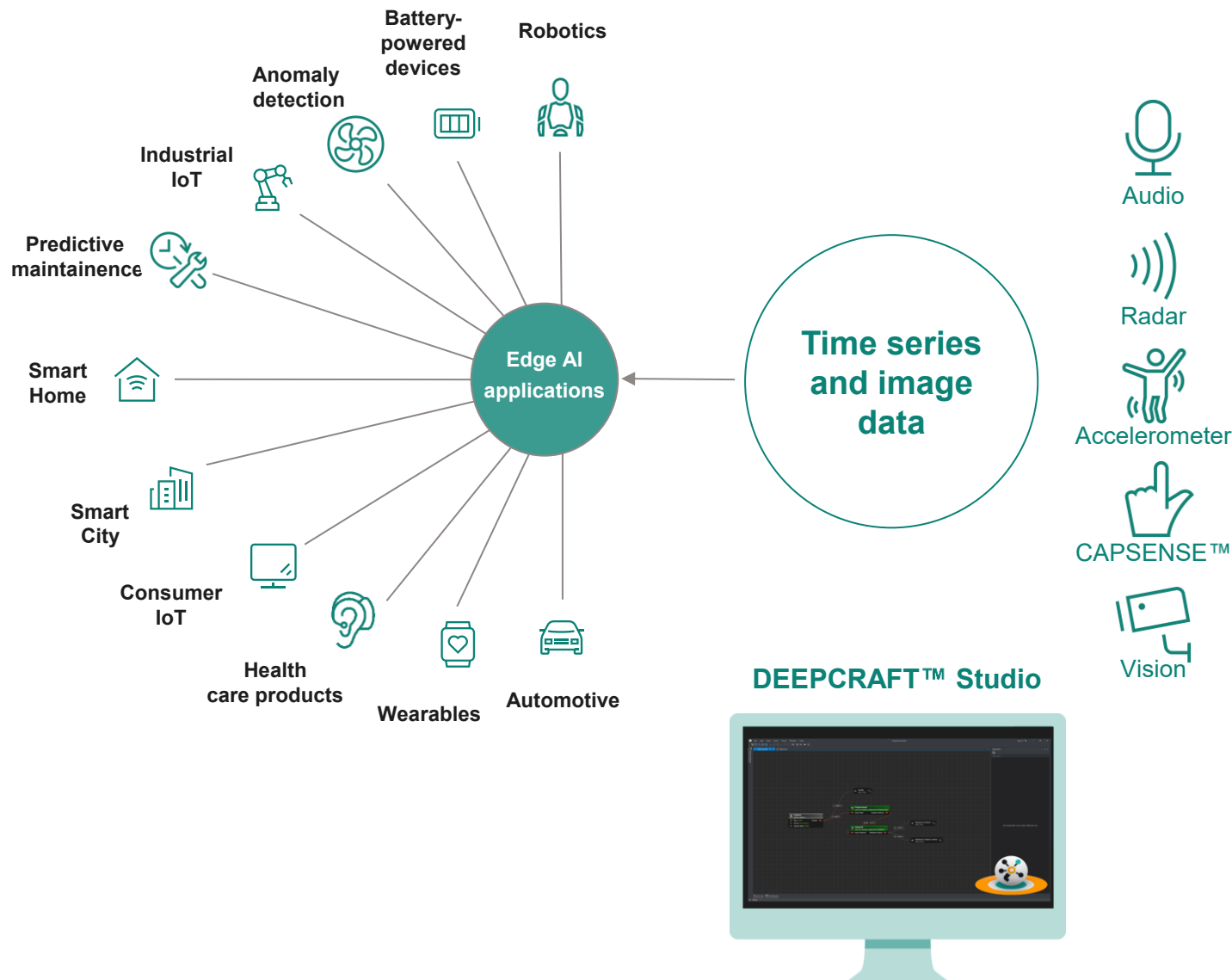


Explore the DEEPCRAFT™ AI Suite,
open-source models, and more.
Everything Edge AI in one place!

DEEPCRAFT™ Studio is an Edge AI development platform

DEEPCRAFT™ Studio

- Train **production-grade custom Edge AI models** with our state-of-the-art platform
- Designed for machine learning (ML) **experts and beginners**
- Leverage pre-trained Studio Accelerators for **shorter time to production**
- Transparent data policy **protects your IP**
- Graph UX interface provides **unique insight** during ML development process
- Focuses on **deep learning**, leveraging modern neural processing units for **enhanced and optimized AI capabilities**



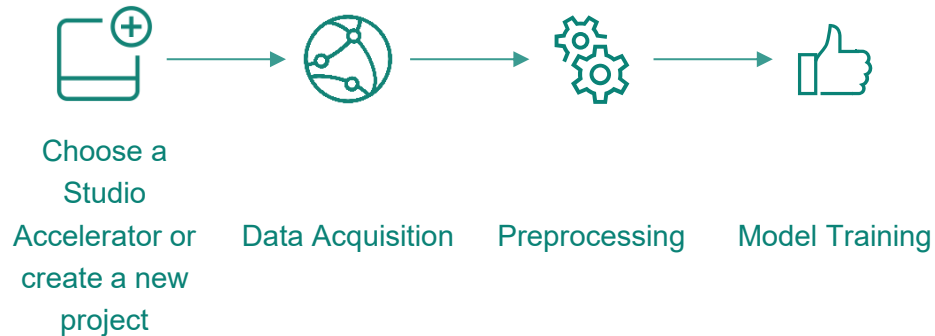
DEEPCRAFT™ Studio provides the full journey from AI/ML model development to embedded software



Build a new custom model, starting from scratch or using a DEEPCRAFT™ Studio Accelerator

Model development and customization

DEEPCRAFT™ Studio



Model optimization

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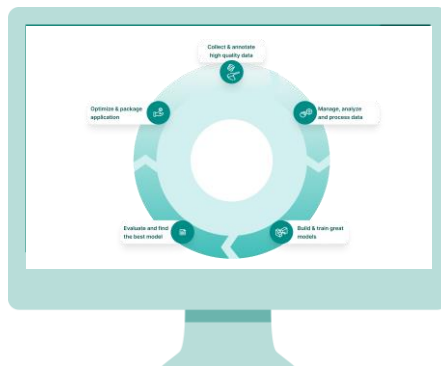


Model deployment on MCU

ModusToolbox™ - for PSOC™ and TRAVEO™



AURIX™ Development Studio



DEEPCRAFT™ Studio Data Collection Live Demo

